

# Lecture 3

## *Microeconomics Review*

Byeong-Hak Choe

*bchoe@geneseo.edu*

*SUNY Geneseo*

August 29, 2025

# Demand, Supply, Equilibrium, and Welfare Analysis

## Demand and Supply Basics

- **Demand Curve (Law of Demand):**  
Quantity demanded decreases as price increases.  
 $Q_d = a - bP, \quad b > 0$
- **Supply Curve (Law of Supply):**  
Quantity supplied increases as price increases.  
 $Q_s = c + dP, \quad d > 0$

## Marginal Benefit (MB) and Demand — Why Downward?

- **MB = WTP:** The demand curve is a **maximum willingness-to-pay** curve; it shows the value of the **next** unit to buyers (their **marginal benefit**).
- **Diminishing marginal benefit:** Early units satisfy the most valuable uses; later units go to less urgent uses  $\Rightarrow$  **MB falls as Q rises**  $\Rightarrow$  **downward slope**.
- **Purchase rule:** Buyers add units while  **$MB \geq P$** .
  - Aggregating individuals, to sell more units the price must fall to attract buyers with lower WTP  $\Rightarrow$  a downward-sloping **demand/MB** curve.

## Marginal Cost (MC) and Supply

- **Supply:** the set of output levels a firm is willing to produce at each possible market price.
  - **Selling rule:** A firm produces and sells units as long as  $P \geq MC$ .
- **Supply comes from MC:** In perfect competition, firms are price takers. They choose output where  $P = MC$ , so the **supply curve is the upward-sloping portion of the MC curve (above shutdown)**.
- **Why MC rises with output:**
  - Fixed capacity → adding more output requires using less efficient inputs.
  - Overtime, congestion, and rising input costs make each extra unit **more costly**.
- **Economies of scale at low Q:** MC may fall initially as fixed costs are spread and efficiency improves. But for supply decisions, only the **rising portion of MC** matters, since that is where firms operate.

## Market Equilibrium

- Market clears when:

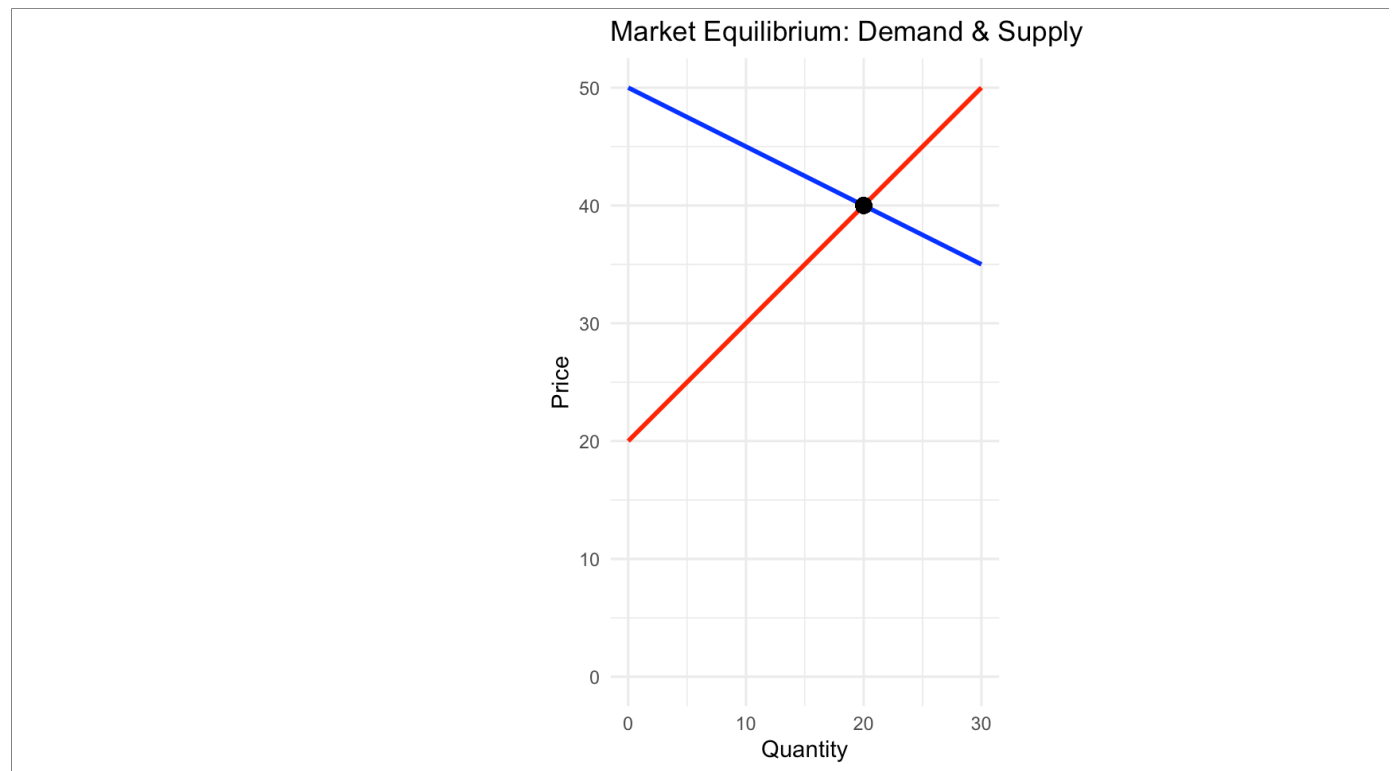
$$Q_d = Q_s$$

- Equilibrium price  $P^*$  is such that market clears ( $Q_d = Q_s$ ).

## Welfare Analysis

- **Consumer surplus (CS):** Area **under demand (MB)** and **above price** up to the purchased quantity.
- **Producer surplus (PS):** Area **above supply (MC)** and **below price** up to the sold quantity.
- **Total Surplus (TS) = CS + PS**
  - TS is the economy's **social welfare** measure.
- At perfect competitive equilibrium:
  - No surplus, no shortage.
  - Welfare is maximized—no mutually beneficial trades are left; any quantity above/below  $Q^*$  reduces TS.

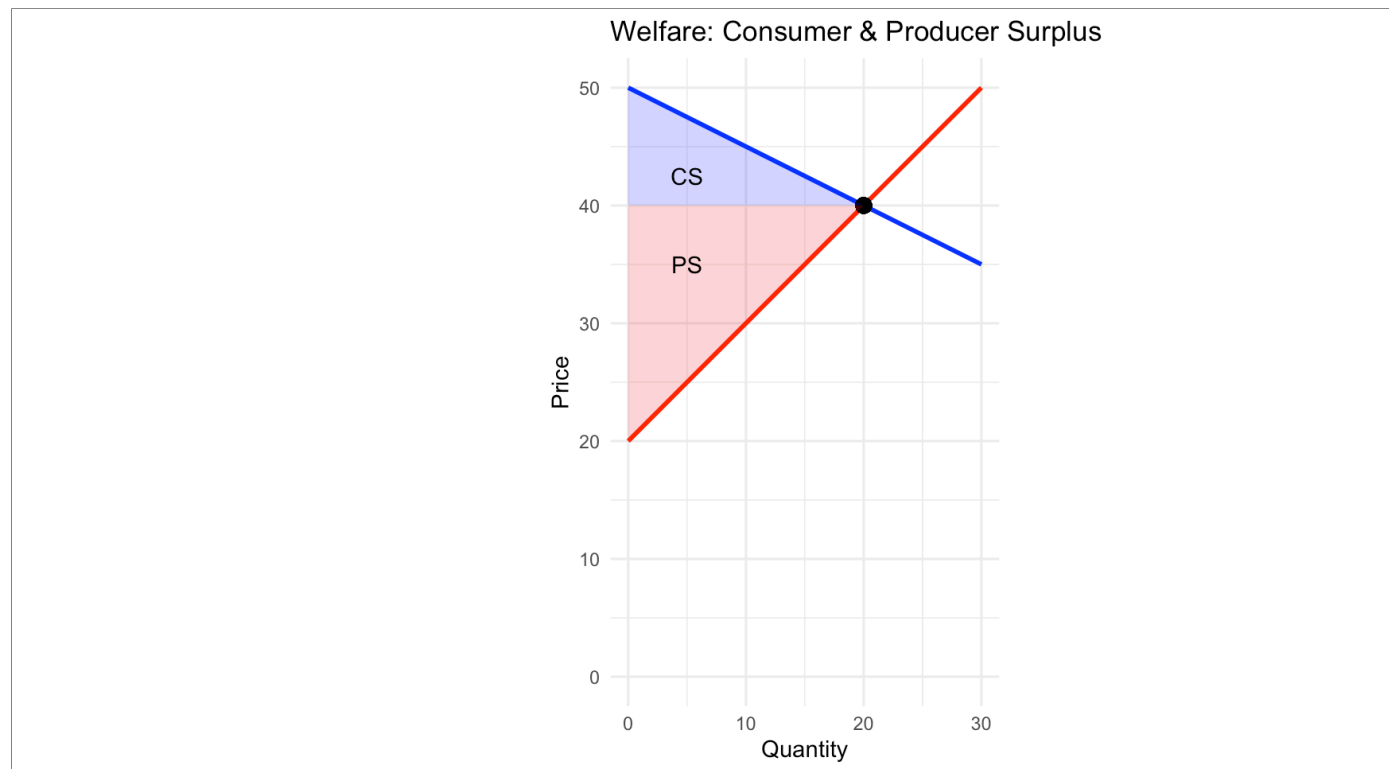
## Demand, Supply, Equilibrium



- What are the equations for the demand and supply curves?



# Consumer and Producer Surplus



- What are the sizes of CS and PS?

## Quantity Regulation (Quota)

- Suppose the government sets a **quota** at  $\bar{Q} = 10$  so that only 10 units can be traded.
- At this restricted quantity, suppose that buyers pay MB at  $Q = 10$  to sellers.
  - How large are **consumer surplus (CS)** and **producer surplus (PS)**?
  - What is the new **total surplus (TS = CS + PS)**, and how does it compare to the no-quota benchmark?

## Tax Example: Setup

- Suppose a per-unit tax  $t = 5$  is imposed on sellers.
- Effect: Supply curve shifts upward by \$5.
  - How large are **consumer surplus (CS)** and **producer surplus (PS)**?
  - What is the new **total surplus (TS = CS + PS + Tax Revenue)**, and how does it compare to the no-tax benchmark?
- **Reminder:** Tax Revenue  $= t \times Q^{**}$ , where  $Q^{**}$  is the quantity traded under the tax.

## Deadweight Loss (DWL)

- **Deadweight loss:** the reduction in **total surplus** that occurs when a market is prevented from reaching the efficient equilibrium quantity.
- **Interpretation:**
  - Represents the value of **mutually beneficial trades that do not occur**.
  - Caused by distortions such as **taxes, quotas, subsidies, or price controls** in a perfectly competitive market.
- **Deadweight loss** is the welfare lost to society when output is restricted away from the point where MB equals MC.